

KERMANSHAH, Iran, Islamic Republic ofWMO: **407660**Lat: **34.3516N**Lon: **47.1534E**Elev: **1319**StdP: **86.45**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-5.1	-18.8	0.8	20.4	-15.2	1.2	15.5	10.1	7.1	8.8	6.9	1.4	100	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	40.1	16.9	39.0	16.7	37.9	16.3	19.7	36.8	18.6	35.8	17.5	34.7	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	10.9	25.4	11.6	10.0	23.6	10.6	9.3	21.8	62.4	36.7	58.2	35.8	54.5	34.6	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.0	7.0	-10.6	42.1	2.9	0.9	-12.7	42.7	-14.4	43.3	-16.0	43.8	-18.1	44.4
			-11.3	19.2	2.6	1.7	-13.2	20.5	-14.7	21.5	-16.2	22.5	-18.1	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	3.2	5.0	9.3	13.6	18.9	24.9	28.6	28.2	23.1	17.4	9.8	5.3
	DBStd	9.34	3.82	3.78	3.55	3.20	2.90	2.33	1.94	1.87	2.36	3.00	3.27	3.21
	HDD10.0	607	213	142	55	6	0	0	0	0	0	0	42	149
	HDD18.3	2014	471	372	282	146	27	0	0	0	1	54	257	404
	CDD10.0	2678	1	3	32	113	277	448	578	565	394	228	36	3
	CDD18.3	1043	0	0	1	3	46	198	320	307	145	24	0	0
	CDH23.3	17277	0	0	14	79	882	3217	5228	4987	2354	513	3	0
	CDH26.7	10251	0	0	1	14	305	1863	3433	3252	1251	133	0	0
(14) Wind	WSAvg	2.7	2.5	2.8	3.2	3.1	3.1	3.1	3.1	2.9	2.6	2.4	2.1	2.1
(15) Precipitation	PrecAvg	418	66	59	63	64	21	1	1	0	1	31	55	55
	PrecMax	735	127	124	160	136	86	8	11	7	16	138	297	149
	PrecMin	271	16	16	2	14	0	0	0	0	0	0	0	3
	PrecStd	98	29	26	42	38	22	2	2	1	4	36	57	33
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	18.1	25.1	28.6	33.9	39.1	41.9	41.0	37.8	31.2	23.0	19.0
		MCWB	6.1	7.8	10.4	13.1	15.4	16.2	17.1	16.8	15.5	13.5	9.8	8.1
	2%	DB	13.2	16.0	21.8	25.2	31.7	37.7	40.5	39.9	36.0	29.9	20.8	15.3
		MCWB	5.7	6.6	9.3	12.5	14.3	15.6	16.8	16.9	14.6	13.2	9.6	6.9
	5%	DB	11.5	14.1	19.1	23.2	29.9	36.1	39.2	39.0	34.4	28.1	18.9	13.2
		MCWB	5.1	6.0	8.2	11.3	13.8	15.3	16.7	16.7	14.2	12.6	8.8	6.2
	10%	DB	9.8	12.2	17.1	21.7	28.1	34.8	38.1	37.9	33.0	26.4	16.9	11.6
		MCWB	4.4	5.3	7.4	10.9	13.5	15.2	16.6	16.3	13.9	12.1	8.4	5.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	9.9	11.6	14.7	17.2	19.6	22.7	21.6	18.5	15.5	12.9	10.0
		MCDB	12.1	14.5	20.7	23.8	30.1	35.1	38.5	38.6	33.8	28.2	18.3	15.1
	2%	WB	7.4	8.1	10.3	13.7	16.0	18.2	20.1	19.7	16.5	14.4	11.5	8.4
		MCDB	10.5	13.6	19.1	22.0	27.9	33.9	37.2	37.4	32.7	26.8	16.3	13.1
	5%	WB	6.0	6.9	9.2	12.7	15.2	17.0	18.8	18.4	15.4	13.5	10.5	7.5
		MCDB	9.7	12.1	16.8	20.6	26.2	32.8	36.0	36.5	31.7	25.0	15.6	11.7
	10%	WB	5.0	5.9	8.2	11.9	14.4	15.8	17.6	17.3	14.5	12.7	9.7	6.4
		MCDB	8.8	10.8	15.2	19.2	25.0	32.4	35.2	35.3	30.8	23.6	14.8	10.3
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.7	13.1	13.9	16.5	19.3	19.3	19.8	19.7	16.5	13.1	11.5
		MCDBR	13.6	15.4	16.8	17.1	19.4	21.0	20.7	20.5	21.0	19.0	16.4	14.6
		MCWBR	7.5	7.9	7.9	7.3	7.6	8.2	7.9	8.2	9.0	9.4	8.1	8.1
		MCDBR	10.4	12.4	13.6	14.2	16.6	18.8	18.7	19.8	20.1	16.4	11.5	11.3
	5% WB	MCWBR	6.4	7.0	6.8	6.5	6.9	8.8	9.5	10.1	10.2	8.8	6.1	6.9
(40) Clear-Sky Solar Irradiance	taub		0.312	0.352	0.408	0.488	0.477	0.427	0.464	0.430	0.373	0.403	0.346	0.316
	taud		2.343	2.218	2.051	1.873	1.923	2.056	1.955	2.049	2.201	2.093	2.278	2.361
	Ebn at Noon		890	885	859	806	819	859	824	847	884	813	838	862
	Edh at Noon		97	123	159	199	192	168	184	165	135	138	103	91
(44) All-Sky Solar Radiation	RadAvg		2.75	3.68	4.86	5.79	6.98	8.32	7.76	7.20	6.23	4.47	3.19	2.56
	RadStd		0.18	0.30	0.40	0.52	0.47	0.34	0.27	0.25	0.24	0.34	0.34	0.25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-3.48	+0.56	-1.68	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	+0.71	-1.56	-0.93	N/A	N/A	N/A	N/A

Nomenclature: See separate page